Project Proposal

Course

Introduction to Embedded System.

Project Title

Smart Automated Agricultural Control & Telemetry System with Edge AI and Telegram Integration.

Team Name

AgriTech Innovators

Team Members

No.	Name	ID	Batch
1	Maria Iffat	2023200000366	65
2	Fahad Ahmed	2023200000371	65
3	Mohammad Fahim	2023200000399	65
4	Safayet Hossain Bhuiyan Munna	2023200000648	65

Section 1: Problem Statement and Social Impact

1.1 Problem Statement

Small-to-medium scale agriculture suffers from unpredictable climate shifts, inefficient resource usage, and delayed crop disease diagnosis. Traditional farming relies on manual, infrequent sampling that misses early indicators of over-watering, pH imbalances, and extreme temperature/humidity stress. Unmonitored environmental fluctuations accelerate crop disease outbreaks, causing major yield losses before symptoms are visually apparent.

1.2 Targeted Community

Smallholder farmers, greenhouse operators, community garden managers, and agricultural researchers in resource-constrained regions who lack access to expensive industrial automation.

1.3 Social Good, Wellbeing, and Environmental Benefits

- **Resource Conservation:** Automated irrigation activates the pump relay strictly when soil moisture drops below *cfg_soilMin*, minimizing water waste and nutrient leaching.
- **Food Security & Crop Protection:** On-device Edge AI classifies crop health from pH, temperature, humidity, and soil data to detect disease risks early and prevent crop failure.
- **Farmer Wellbeing & Remote Control:** Bidirectional Telegram commands (`/status`, `/pump_on`, `/auto_on`) reduce manual field labor and enable remote management.
- **Hazard Mitigation:** Automated cooling fans trigger when temperature exceed *`cfg_tempMax`* or humidity surpasses 85%, protecting crops against heat waves.

Section 2: Sustainability and Accessibility

2.1 Sustainability

- **Energy Efficiency:** On-device Edge Impulse ML running on CPU Core 1 eliminates continuous high-power Wi-Fi data streaming. CPU Core 0 activates Wi-Fi only for Telegram polling or alerts.
- **Low Maintenance:** Non-Volatile Storage (NVS via `Preferences.h`) retains crop profiles and alert thresholds across power cycles without manual reconfiguration.
- **Hardware Longevity:** Solid-state relay control ensures high-power actuators run only when needed, reducing wear.

2.2 Accessibility

- **Affordable Hardware:** Uses low-cost components (ESP32, DHT22, capacitive soil sensor) to keep total build costs under Tk 3000.
- **Dual User Interface:**
    - *Online Users:* Remote monitoring, threshold tuning, and manual overrides via Telegram Bot.
    - *Offline Users:* Standalone 1.44" ST7735 TFT display with analog joystick menu navigation allows full offline operation.
- **Inclusive Feedback:** Visual NeoPixel LED patterns and distinct piezo audio frequencies provide non-text alerts for users with low literacy.

Section 3: Team Formation and Task Distribution

3.1 Team Members

- **Mohammad Fahim [ID-399] (Team Leader & Lead Architect):** System Architecture, Edge AI Model Training & Inference, Master Hardware Schematics, FreeRTOS, Telegram Bot.
- **Maria Iffat [ID-366] (Hardware & Sensor System Engineer):** Environmental Sensors (DHT22, Soil, LDR), Analog Signal Calibration, Dual-Relay Actuators & Threshold Logic.
- **Safayet Hossain [ID-648] (UI/UX & Graphic System Engineer):** ST7735 Display Firmware, Custom Bitmaps, Joystick Navigation Engine & NVS Storage.
- **Fahad Ahmed [ID-371] (Integration & Verification Engineer):** Enclosure Build, Assembly, Fault Injection Debugging, System Testing & Documentation.

3.2 Task Distribution

Week	Task / Responsibility	Primary Assigned Member	Deliverables
1	Master hardware schematics, power routing calculations (5V/3.3V), pinout mapping	Mohammad Fahim [ID-399]	Master Circuit Schematics & Power Distribution Design
2	Edge Impulse dataset collection, feature extraction, ML model training, C++ inference deployment (`runAIInference()`)	Mohammad Fahim [ID-399]	Deployed Edge AI Model & Health Classifier
3	Telegram Bot API setup, command parser (`/status`, `/pump_on`), cloud alert engine	Mohammad Fahim [ID-399]	Bidirectional Cloud Telemetry & Remote Control.
4	Dual-core FreeRTOS task scheduling (`xTaskCreatePinnedToCore`) & mutex locking (`telemetryMutex`)	Mohammad Fahim [ID-399]	Thread-Safe Multi-Core Architecture.
5	Sensor array wiring (DHT22, Soil, LDR), analog signal processing, relay control routines.	Maria Iffat [ID-366]	Calibrated Sensor Libraries & Relay Routines.
6	ST7735 SPI display integration, custom status bitmaps (`drawWifiIcon`, `drawAIIcon`), UI card rendering.	Safayet Hossain [ID-648]	Local Graphic Display UI
7	Analog joystick menu navigation, deadzone tuning, NVS storage (`Preferences.h`) routines	Safayet Hossain [ID-648]	Dynamic Joystick UI & Persistent Memory Manager.
8	Physical enclosure build, wire harness assembly, full system integration, cross-core debugging.	Fahad Ahmed [ID-371]	Integrated Prototype & Verification Logs
9	System stress testing, final proposal report, presentation deck, live demonstration setup	Fahad Ahmed & All [ID-371]	Final Documentation & Live Demo Readiness

Section 4: Initial Budget Estimate

4.1 Tabular Budget

Component	Quantity	Unit Price	Total
ESP32 Dev Board	1	380	380
ESP32 Extension Board	1	180	180
ST7735 1.44” SPI Color TFT Display	1	450	450
DHT22 Temp & Humidity Sensor	1	220	220
Capacitive Soil Moisture Sensor	1	60	60
LDR Digital Light Sensor Module	1	40	40
4-Channel 5V Relay Module	1	280	280
5V Submersible Water Pump	1	50	50
5V DC Cooling Fan	1	40	40
WS2812B NeoPixel RGB LED Strip	1	150	150
2-Axis Analog Joystick Module	1	80	80
Active Buzzer	1	20	20
Breadboard, Jumpers & Other	1	1000	1000
Total Estimated Cost:2950			

4.2 Component Justification

- **ESP32 Microcontroller:** Selected over Arduino Uno (Tk 500) or Pico W (Tk 1500) for its dual-core 240MHz CPU necessary to execute FreeRTOS and Edge AI inferencing simultaneously alongside native Wi-Fi.
- **Capacitive Soil Sensor:** Chosen over resistive sensors (TK 50) because capacitive probes avoid soil contact corrosion, ensuring long-term hardware reliability.
- **DHT22 Sensor:** Offers higher accuracy (±2-5%) and wider temperature range (-40 to +80°C) than DHT11 (TK 120), critical for reliable AI risk prediction.
- **ST7735 TFT Display:** Enables dynamic card layouts and color status indicators for offline navigation.
- **Relay Module:** Provides isolation to shield ESP32 GPIO pins from inductive motor voltage spikes.